

ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 8 – PROGRAM DESIGN 5
(SPECIAL POPULATIONS)

- CHAPTER 14
EXERCISE ACROSS THE LIFESPAN



CHAPTER 14

EXERCISE ACROSS THE LIFESPAN

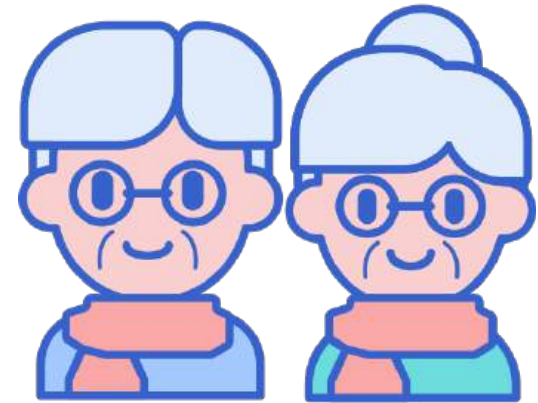
CONTENT COVERED



Exercise for Youth



Exercise for Women
during Pregnancy and
the Postpartum Period



Exercise for Older Adults

EXERCISE FOR YOUTH

Learning Objectives:

- Learn the exercise guidelines for youth
- Apply the ACE IFT Model to youth

EXERCISE AND YOUTH

- Globally, the majority of youths are not getting the recommended amount of physical activity
- 2018 US Report Card on Physical Activity for Children and Youth: **D-**, 20-26% meeting recommendations (World average: **D**)



- Reduced physical activity + poor dietary habits $\Rightarrow$ weight gain:
 - Childhood obesity, hypertension, type 2 diabetes, osteoporosis, atherosclerosis
 - Increased sedentary behavior into adulthood, $\nearrow$ risk of premature death

PROGRAM CONSIDERATIONS

- GENERAL

- Total amount of physical activity accumulated is more important
- Bone strengthening activities
- Recreational and competitive sports
- Careful consideration: overweight, obesity, less developed in motor skills
- Fun and appealing to the child's unique interest
- Encourage experimentation with new activities
- Muscle strengthening activities incorporated into play and games



PROGRAM CONSIDERATIONS - SAFETY

Risk factors for children:

- Epiphyseal plate (growth plate) not fully fused

Guidelines:

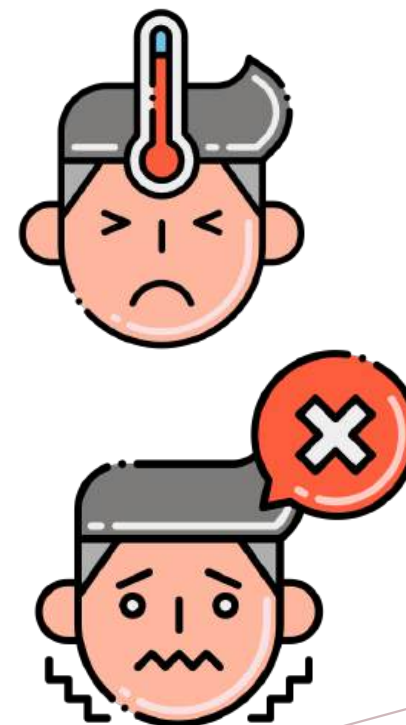
- Ensure proper supervision
- Emphasize proper technique/forms at all times
- FW/BW exercises (traditional machine might not be suitable)
- NO 1RM lifts, no sudden explosive moves, or compete with other children while performing muscular training
- Teach proper breathing during exercise
- Encourage hydration
- Encourage communication with PT, coach, parent, or teacher
- Variety, fun, periodization to avoid overuse injury



PROGRAM CONSIDERATIONS - ENVIRONMENT

Greater risks compared to adults

- In the Heat:
 - Higher surface area to mass ratio
 - Lower exercise economy
 - Diminished sweating capacity
 - Lower cardiac output at a similar workload
- In the Cold:
 - Dehydration
 - Hypothermia
 - Frostbite
 - Dry air exacerbates asthma or exercise-induced bronchoconstriction



FITT RECOMMENDATIONS FOR YOUTH

	Aerobic	Resistance	Bone Strengthening
Frequency	Daily	≥3 days/week	≥3 days/week
Intensity	Most should be moderate (noticeable increase in HR and breathing) to vigorous intensity (substantial increase in HR and breathing). Include vigorous intensity at least 3 days/week.*	Use of body weight as resistance or 8–15 submaximal repetitions of an exercise to the point of moderate fatigue with good mechanical form	N/A
Time	As part of ≥60 minutes/day of exercise	As part of ≥60 minutes/day of exercise	As part of ≥60 minutes/day of exercise
Type	Enjoyable and developmentally appropriate activities, including running, brisk walking, swimming, dancing, bicycling, and sports such as soccer, basketball, or tennis	Muscle-strengthening physical activities can be unstructured (e.g., playing on playground equipment, climbing trees, tug of war) or structured (e.g., lifting weights, working with resistance bands)	Bone-strengthening activities include running, jump rope, basketball, tennis, resistance training, and hopscotch

Table 14-2: FITT Recommendations for Children and Adolescents

EXERCISE FOR WOMEN DURING PREGNANCY & POSTPARTUM PERIOD

Learning Objectives:

- Identify the physical and physiological changes during pregnancy
- Know exercise guidelines for pregnant women
- Apply ACE-IFT Model for pregnant women
- Know postpartum exercise guidelines

EXERCISE FOR PREGNANT & POSTPARTUM WOMEN

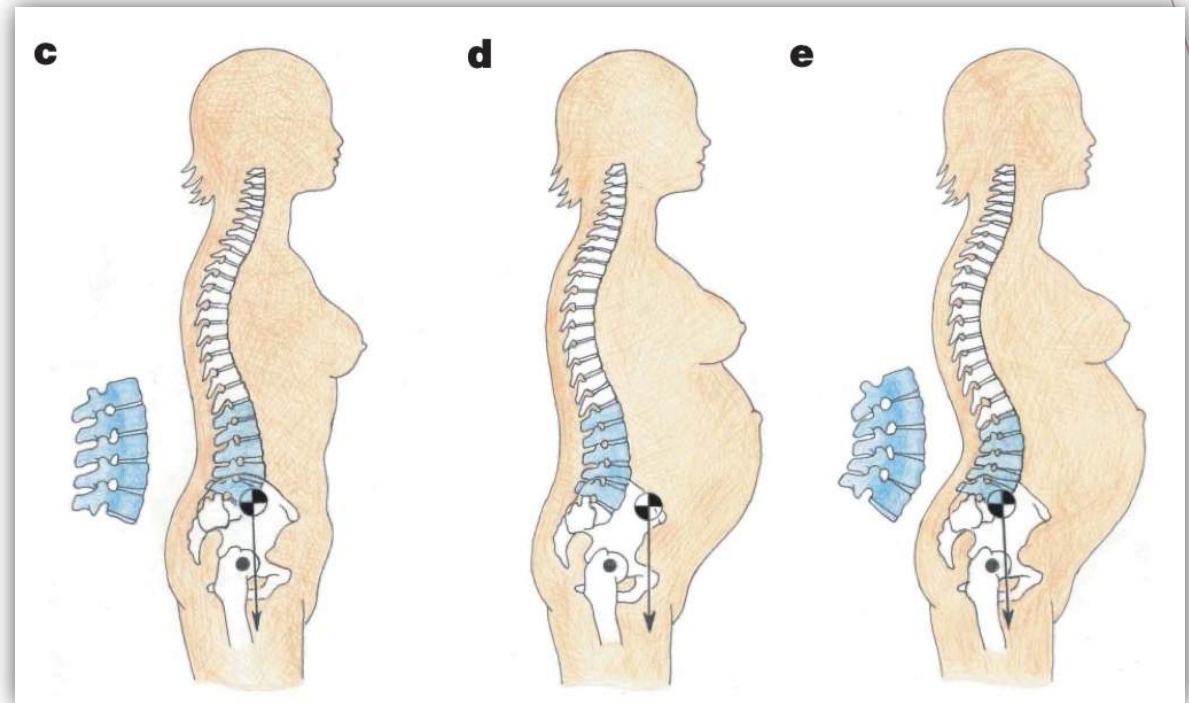
- The beneficial effects of exercise in most adults...far outweigh the risks (ACOG, 2015)
- Regular exercise is associated with reduced rates of preeclampsia, gestational diabetes mellitus, Caesarean, low-back pain, anxiety, nausea, heartburn, insomnia, leg cramps, and excessive weight gain
- Only 15 percent of women meet the recommended guidelines for exercise during pregnancy.*
- Medical clearance required for women with other diseases



*Source from H. Prather, T. Spitznagle, and D. Hunt, *PM&R*, vol. 4, no. 11, pp. 845–850, Nov. 2012, doi: [10.1016/j.pmrj.2012.07.012](https://doi.org/10.1016/j.pmrj.2012.07.012).

PHYSICAL & PHYSIOLOGICAL CHANGES DURING PREGNANCY

- Healthy weight gain 25-35 lb
- Increased abdominal circumference
- COG moves upward and forward
- Stress on back, pelvis, hips, and legs
- Compromised balance and coordination



Lumbar lordosis during pregnancy

K. K. Whitcome, L. J. Shapiro, and D. E. Lieberman, *Nature*, vol. 450, no. 7172, pp. 1075–1078, Dec. 2007.

PHYSICAL & PHYSIOLOGICAL CHANGES DURING PREGNANCY

- Healthy weight gain 25-35 lb with variations
- Increased abdominal circumference
- COG moves upward and forward
- Stress on back, pelvis, hips, and legs
- Compromised balance and coordination
- Change in hormone levels $\Rightarrow$ nausea, fatigue
- $\nearrow$ Relaxing hormone $\Rightarrow$ $\nearrow$ joint laxity $\Rightarrow$ joint instability
- $\nearrow$ HR, blood volume, SV, Q at rest; $\searrow$ Cardiac reserve; .
More difficulty exercising
- $\nearrow$ Tidal volume, O_2 consumption
- $\nearrow$ Heat dissipation

$\nearrow$ Posterior leg / trunk strength

Adjust intensity

Strength & stability training

$\nearrow$ rest periods

Watch out T & RH

EXERCISE GUIDELINES - PREGNANCY

- Previously active women with no health complication may continue their previous level of exercise as tolerated
- Previously inactive women should begin low and slowly progress to moderate intensity exercise
- Use talk test and stay below VT1
- Focus on hydration and balancing caloric intake; some may take a small snack prior to exercise to avoid hypoglycemia

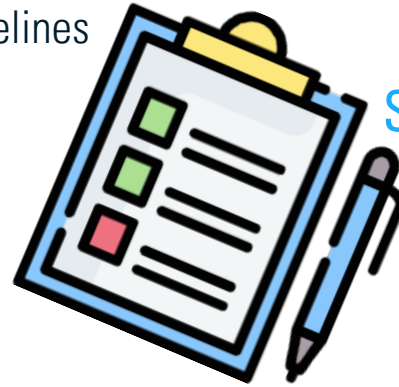
Avoid:

- Activities that risk falling
- Contact sports
- Extreme heat
- Prolonged supine position after about 20 weeks of gestation (risk of hypotension, ie., low blood pressure)



PROGRAM CONSIDERATIONS - SAFETY

- Follow updated ACOG guidelines
- Use PARmed-X form



Contraindications:

- Risk factors for pre-term labor
- Vaginal bleeding
- Premature rupture of membranes
- Severe anemia

Signs to terminate exercise session:

- Vaginal bleeding
- Regular painful contractions
- Amniotic fluid leakage
- Dyspnea before exertion
- Dizziness or feeling faint
- Headache
- Chest pain
- Muscle weakness affecting balance
- Calf pain or swelling



FITT RECOMMENDATIONS - PREGNANCY

	Aerobic	Resistance	Flexibility
Frequency	≥3–5 days/week	2–3 nonconsecutive days/week	≥2–3 days/week, with daily being most effective
Intensity	Moderate intensity (3–5.9 METs; RPE of 12–13 on the 6–20 scale); vigorous-intensity exercise (≥6 METs; RPE 14–17 on the 6–20 scale) for women who were highly active prior to pregnancy or for women who progress to higher fitness levels during pregnancy*	Intensity that permits multiple submaximal repetitions (i.e., 8–10 or 12–15 repetitions) to be performed to a point of moderate fatigue	Stretch to the point of feeling tightness or slight discomfort
Time	~30 minutes/day of accumulated moderate-intensity exercise to total at least 150 minutes/week or 75 minutes/week of vigorous-intensity aerobic exercise	One set for beginners; two to three sets for intermediate and advanced; target major muscle groups	Hold static stretch for 10–30 seconds
Type	A variety of weight- and non-weight-bearing activities are well tolerated during pregnancy (e.g., hiking, group exercise, swimming).	A variety of machines, free weights, and body-weight exercises are well tolerated during pregnancy (e.g., upright chest press, dumbbells, lunges).	A series of static (i.e., active or passive) and dynamic flexibility exercises for each muscle-tendon unit.

Table 14-3: FITT Recommendations for Women Who are Pregnant

EXERCISE GUIDELINES - POSTPARTUM

- Obtain physician clearance and/or guidelines if medical or surgical complications are present
- Begin slow, gradually ↗ duration, frequency, then intensity
- Slower return to exercise after C-section
- 20-30 min daily activity. E.g. walking, simple muscular exercises
- Optimal hydration and healthy eating; account for energy needs for physical activity and lactation
- Stop if: unusual pain, unusual bright red vaginal bleeding



EXERCISE GUIDELINES - POSTPARTUM

- Back and posterior shoulder exercises to correct forward rounded shoulders due to breastfeeding/sitting with a child for long periods of time
- Promote trunk strength and stability
- Take into consideration of a healing C-section incision, presence of diastasis recti



ADDITIONAL READING MATERIALS

[1] ACOG, “Physical Activity and Exercise During Pregnancy and the Postpartum Period”, Apr 2020. <https://www.acog.org/clinical/clinical-guidance/committee-opinion/articles/2020/04/physical-activity-and-exercise-during-pregnancy-and-the-postpartum-period>

EXERCISE AND OLDER ADULTS

Learning Objectives:

- Identify the structural changes with aging
- Understand the importance of stability and balance for older adults
- Apply ACE-IFT Model for older adults

OLDER ADULTS

Signs of aging:

- Decline in fitness
- Loss of height
- Reduced lean body mass
- Loss of skin/connective tissue elasticity
- Slower healing
- Changes in eyesight
- Reduced coordination

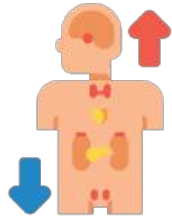
Mental health

- Cognitive impairment
- Increase in depression and anxiety disorder
- Loss of social stimulation

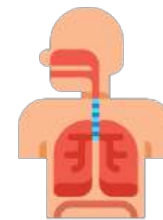
Noticeable changes in various systems:



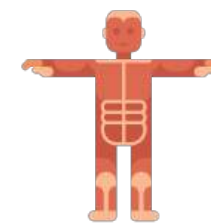
Cardiovascular



Endocrine



Respiratory



Musculoskeletal

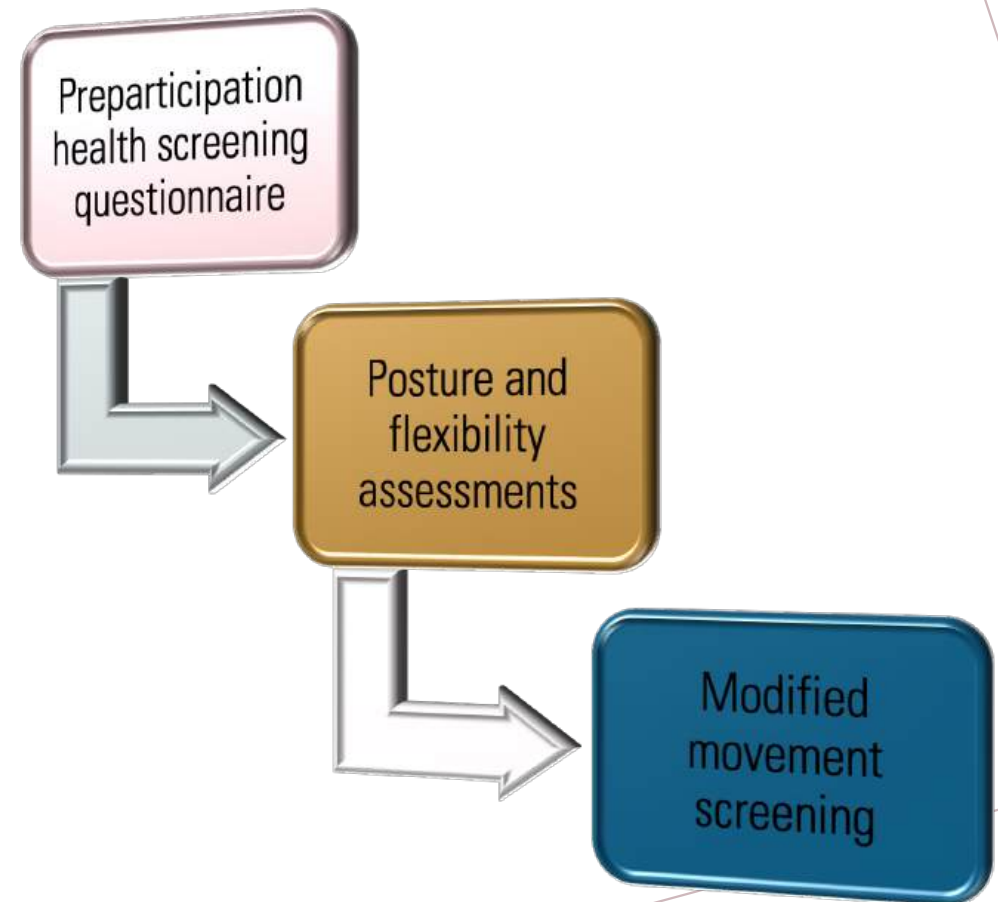


BENEFITS OF EXERCISE FOR OLDER ADULTS

- 1.
- 2.
- 3.

EXERCISE GUIDELINES – OLDER ADULTS

- Similar guidelines to those for general population
- Pay attention to chronic diseases, e.g. arthritis
- Movement assessments may need to be modified
- Balance training with support
- Functional training
- Weight-bearing CT
- Prolonged warm-up & cool-down
- Goal setting



Balance exercises and training progressions for older adults

Position	Balance exercise
Sitting	-Sit upright and complete progressions listed below. -Perform leg activities (heel, toe, or single-leg raises, marching).
Standing	-“Clock” – balance on one leg (other leg at 45° or 90°), personal trainer calls out time, client moves nonsupport leg to the time called out, alternate legs. -Performing leg activities (heel, toe, or single-leg raises - 45° or 90° angle, marching). -Spelling – balance on one leg, personal trainer asks the client to spell word working with nonsupport leg (i.e., client’s name, day of week, favorite food), alternate leg.
In-motion	-Heel-to-toe walking along 15-ft line on floor (first with and then without partner). -“Excursion” – alternating legs, lunge over a space separated by two lines of tape. Progress to hopping or jumping (using single-leg or double-leg) back and forth across the space. -Dribble basketball around cones that require the client to change direction multiple times.
Training progression	-Arm progression: use surface for support, hands on thigh, hands folded across chest. -Surface progressions: chair, balance disc, foam pad, physioball. -Visual progressions: open eyes, sun glasses or dim room lighting, closed eyes. -Task progression: single tasking, multi-tasking (i.e., balance exercise + pass/catch ball)

BALANCE EXERCISE & TRAINING PROGRESSIONS FOR OLDER ADULTS

FITT RECOMMENDATIONS FOR OLDER ADULTS

	Aerobic	Resistance	Flexibility
Frequency	≥5 days/week for moderate intensity; ≥3 days/week for vigorous intensity; 3–5 days/week for a combination of moderate and vigorous intensity	≥2 days/week	≥2 days/week
Intensity	On a scale of 0–10 for level of physical exertion, 5–6 for moderate intensity and 7–8 for vigorous intensity*	Light intensity (i.e., 40–50% 1-RM) for beginners; progress to moderate-to-vigorous intensity (60–80% 1-RM); alternatively, moderate (5–6) to vigorous (7–8) intensity on a 0–10 scale	Stretch to the point of feeling tightness or slight discomfort.

Table 14-4: FITT Recommendations for Older Adults

Time	30–60 minutes/day of moderate-intensity exercise; 20–30 minutes/day of vigorous-intensity exercise; or an equivalent combination of moderate- and vigorous-intensity exercise; may be accumulated in bouts of at least 10 minutes each	8–10 exercise involving the major muscle groups; 1–3 sets of 8–12 repetitions each	Hold stretch for 30–60 seconds.
Type	Any modality that does not impose excessive orthopedic stress such as walking. Aquatic exercise and stationary cycle exercise may be advantageous for those with limited tolerance for weight-bearing activity.	Progressive weight-training programs or weight-bearing calisthenics, stair climbing, and other strengthening activities that use the major muscle groups	Any physical activities that maintain or increase flexibility using slow movements that terminate in static stretches for each muscle group rather than rapid ballistic movements

Table 14-4: FITT Recommendations for Older Adults (con't)